A Novel CFAR-Based Ship Detection Method Using Range-Compressed Data for Spaceborne SAR System

Chao Wang, Baolong Guo, *Senior Member, IEEE*, Jiawei Song, Fangliang He, and Cheng Li

***Abstract*— Spaceborne synthetic aperture radar (SAR) image ship detection is an important tool to ensure the safety of sea areas and improve the efficiency of maritime traffic. Due to the sparse distribution of ships in the vast ocean, many imaging results are redundant. Furthermore, SAR imaging consumes huge computing, storage, and communication resources. The range-compressed data, without azimuth compression calculation, has caught our attention. Nevertheless, the echo energy of the ship is scattered in the azimuth direction, making it difficult to detect. Several deep learning-based methods are proposed, yet their performance is constrained by labeled datasets. As they neglect sea clutter interference, these methods are also impractical. To address these issues, this article proposes a constant false alarm rate (CFAR)-based ship detector for range-compressed SAR data. First, the imaging process of spaceborne SAR signal is reviewed and analyzed. Then, a generalized Gamma distribution (GΓD)-based sea clutter model is proposed for the SAR range-focused domain. Next, a CFAR-based method for detecting ships in range-compressed SAR data is customized. Finally, experiments are conducted on Sentinel-1 and ERS-2 SAR data. The results show that the proposed sea clutter model has high goodness-of-fit, and the customized CFAR-based method effectively detects ship targets. In summary, ship detection in range-compressed SAR data is very promising research.**

***Index Terms*— Range-compressed data, sea clutter, ship detection, synthetic aperture radar (SAR).**

I. Introduction

SPACEBORNE synthetic aperture radar (SAR) plays an important role in maritime ship monitoring [1]. With the development of remote sensing technology and the increase in demand, SAR systems exhibit higher spatial resolution and wider scanning widths. This implies that spaceborne SAR ship detectors will face greater challenges in real-time and effectiveness. At present, the research object of traditional SAR ship detection is SAR images, such as single-look complex (SLC) and multilook images. Constant false alarm rate (CFAR)-based methods detect pixels with higher intensity based on the statistical characteristics of background clutter in images [2], [3], [4]. Machine-learning-based methods are devoted to exploring the visual features of ships in images [5], [6], [7]. The sensing scene of spaceborne SAR over the sea is vast, with ships sparsely distributed. The large areas in SAR images are redundant, which wastes computing resources. Moreover, the SAR imaging process is quite time-consuming, which becomes the bottleneck for the speed of ship detection.

Recently, ship detection before SAR imaging has attracted wide attention. Leng et al. [8] propose a new method to detect ships using raw SAR echo data. However, ship and sea clutter scattering are difficult to distinguish in the raw data, which makes it unsuitable for complex sea conditions. To tackle this issue, methods of sensing ships from range-compressed SAR data are proposed. Joshi and Baumgartner [9], [10], Joshi et al. [11] propose a series of methods for ship detection in the range-Doppler domain. But, such methods require azimuth Fourier transform of the echo data, which consumes a lot of computational resources. Based on convolutional neural networks, some studies propose to detect ships directly from range-compressed SAR data [12], [13], [14], [15]. However, these methods require a large number of labeled datasets, which is difficult to prepare. In addition, the computational complexity of these networks is high. Nonetheless, detecting ships in range-compressed spaceborne SAR data offers the following two advantages: 1) no azimuth compression calculation of the vast ocean background, which saves computing resources and 2) only the range-compressed data of the ship is preserved, which saves storage and communication resources.

Hence, this article proposes to apply CFAR to range-compressed SAR data. The accuracy of CFAR is closely related to sea clutter modeling. Log-normal [16] and Weibull [17] are simple empirical models that require substantial data support and are prone to overfitting. Considering the thermal noise of the system, K + Noise [18] and K + Rayleigh [19] distributions are proposed. However, these models are unstable and time-consuming. The trimodal discrete texture (3MD) model uniquely models the sea clutter texture as discrete components [20], [21], [22]. Nonetheless, its mechanism of discretization renders it unsuitable for clutter simulation. Experiments with the generalized Gamma distribution (GΓD) model, as shown in [23], [24], [25], [26], and [27], demonstrate superior fitting performance. Additionally, the GΓD model features a closed-form threshold expression, significantly facilitating its application [28]. Therefore, this article

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chooses the GΓD model to explore the sea clutter fitting in the range-compressed SAR data.

The characteristics of SAR echo signal in range-focused domain and image domain are different. Thus, the clutter model, which is designed for images, cannot be directly used in range-compressed data. Based on probability theory and SAR imaging mechanism, a clutter model suitable for range-compressed SAR data needs to be derived. However, several critical issues remain to be addressed:

1) *Lack of Modeling Experience:* Up to now, there is no research on sea clutter modeling in range-compressed SAR data.
2) *High Computational Complexity:* For the GΓD model, parameter estimation involves numerical iteration, which brings a heavy computational burden [26].
3) *Strong Sea Spike Interference:* Due to the unfocused azimuth energy, it is difficult to distinguish high-intensity pixels generated by sea spikes [29].

To address these issues, this work comprehensively explores the characteristics of range-compressed SAR data. The sea clutter distribution model, parameter estimation method, and ship detection strategy in the range-focused domain are customized. To be specific, this article proposes a CFAR-based method to detect ships from range-compressed SAR data. The main contributions of this article are as follows:

1) For range-compressed SAR data, a sea clutter model based on GΓD is proposed, which is called RC-GΓD. Based on the SAR imaging mechanism and probability theory, the applicability of the model is theoretically proved. Meanwhile, considering the estimation precision and computational complexity, an analytical scheme of parameter estimation is given.
2) Combined with the RC-GΓD model, a new CFAR-based detector is proposed, which is called RC-GΓD-CFAR. Considering the scattering signal distribution in range-compressed data, a nonlocal clutter data sampling method is present. Then, a noniterative method for adaptive threshold calculation is also given. To eliminate the false alarm caused by sea spikes, a 1-D ship extractor is customized.
3) Experiments are conducted on the data of Sentinel-1 and ERS-2 in different regions, which proves that the RC-GΓD model has good fitting accuracy. It is also proved that RC-GΓD-CFAR can successfully detect ship targets from range-compressed SAR data. In addition, azimuth compressing the range-compressed data of the target area can obtain clearer ship images. To the best of our knowledge, this is the first time to propose a CFAR-based ship detector for spaceborne range-compressed SAR data.

The rest of this article is organized as follows. In Section II, the imaging process of the raw echo data of spaceborne SAR is analyzed. Based on the analysis, the RC-GΓD model is derived in Section III, and its parameter estimation method is given. Then, in Section IV, the RC-GΓD-CFAR ship detection method is proposed. The experimental results based on real SAR data are presented and analyzed in Section V. Subsequently, Section VI discusses the significance of using CFAR and the application prospect of this study. Finally, Section VII concludes this article.

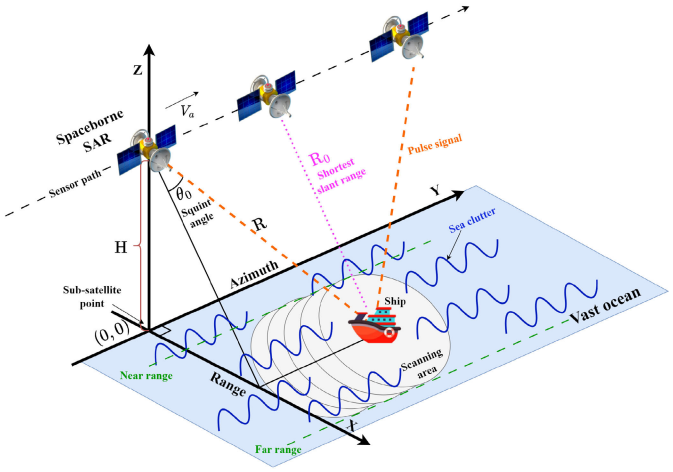


Fig. 1. Marine ship data acquisition geometry of the side-looking mode spaceborne SAR. The scene is dominated by the ocean, including a ship target and a large number of sea clutter.

II. Preliminaries

In this section, we mainly review the SAR imaging mechanism and explore the distribution of energy in the range-focused domain. First, the mathematics of spaceborne SAR echo signal imaging is reviewed. Next, the feasibility of detecting ships from range-compressed data is analyzed. Finally, an overview of the working paradigm of the proposed method is elaborated.

A. SAR Echo Signal and Imaging Method

Considering the spaceborne SAR system, the geometry of the side-looking mode with a squint angle of θ_0 is shown in Fig. 1. The satellite moves along the Y-axis, called the azimuth direction, at a constant speed of V_a and a height of H. The X-axis perpendicular to the Y-axis in the same horizontal plane represents the range direction. The original slant range between the ship center and the platform trajectory along the squint direction is R, and the minimum value is R_0. The pulse signal transmitted by SAR is usually a linear frequency modulation signal [30]

$$s(t_r) = \mathrm{rect}\left(\frac{t_r}{T_p}\right) \cdot \exp\left\{j2\pi f_c t_r + j\pi k_r {t_r}^2\right\} \tag{1}$$

where t_r is the range fast time, T_p is the width of the transmitted pulse, rect$(\cdot)$ is the rectangular window of the signal, f_c is the carrier frequency, and k_r is the range frequency.

Let t_a denotes the azimuth slow time of spaceborne SAR, the baseband form of the echo signal can be expressed as [8]

$$s(t_r, t_a) = \sigma_s \cdot w_r\left(t_r - \frac{2R(t_a)}{c}\right) \cdot w_a(t_a) \cdot \exp\left(-\frac{j4\pi f_c R(t_a)}{c} + j\pi k_r \left(t_r - \frac{2R(t_a)}{c}\right)^2\right) \tag{2}$$

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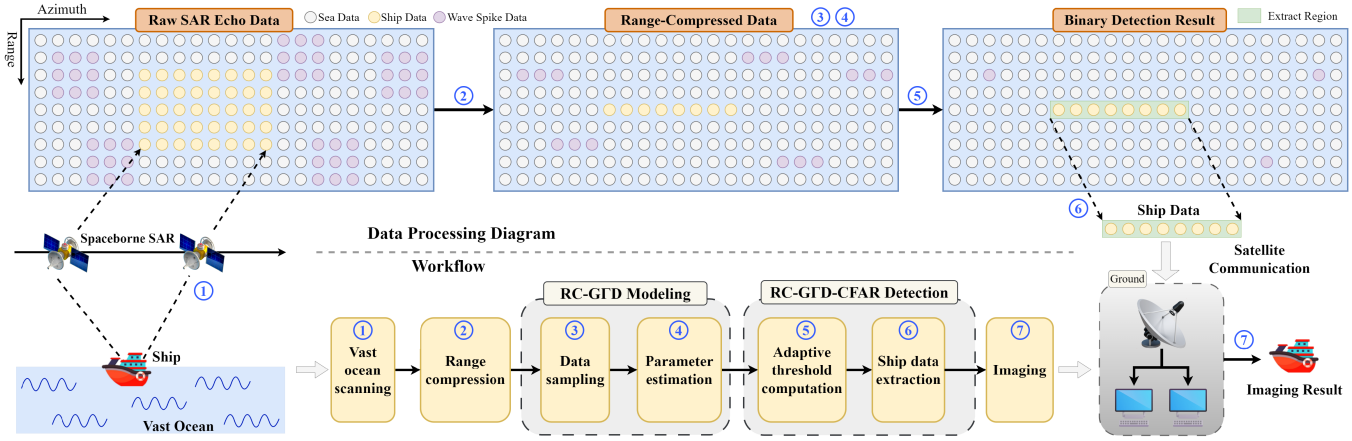


Fig. 2. Overview of the working paradigm of detecting ships from range-compressed SAR data. The upper part is the processing diagram of SAR echo data. The lower part is the functional workflow. Paired serial numbers represent the functional modules used in data processing.

where σ_s is the backscattering coefficient of the ship target, $w_r(\cdot)$ is the range envelope function, $w_a(\cdot)$ is the azimuth envelope function, c is the speed of light, and $R(t_a)$ is the distance between the target and the radar.

Spaceborne SAR systems, such as ERS or Sentinel-1, are typically low squint and medium aperture lengths. Therefore, the beam direction is close to the zero Doppler direction, which means that the distance $R(t_a)$ can be approximated as a parabolic form [31]

$$R(t_a) = \sqrt{R_0^2 + V_a^2 {t_a}^2} \approx R_0 + \frac{V_a^2 {t_a}^2}{2R_0}. \tag{3}$$

Usually, the H of low-orbit SAR satellites is greater than 200 km, $V_a \approx 7.9$ km/s, and $t_a \leq 1 \times 10^{-6}$ s. As for geosynchronous satellites, $H = 3.6 \times 10^4$ km, $V_a \leq 7.0$ km/s, and $t_a \leq 1 \times 10^{-3}$ s. Hence, in spaceborne SAR imaging, there is a hidden condition $2R_0 > 2H \gg V_a^2 {t_a}^2$, so that (3) can be further simplified as $R(t_a) \approx R_0$.

Then, the echo signal after range compressing can be derived as follows:

$$s_{rc}(t_r, t_a) = \sigma_s \cdot w_a(t_a) \cdot \mathrm{sinc}\left[B_r\left(t_r - \frac{2R_0}{c}\right)\right] \cdot \exp\left(-\frac{j4\pi f_c R_0}{c}\right) \cdot \exp\left(-\frac{j2\pi f_c V_a^2 t_a^2}{cR_0}\right). \tag{4}$$

It can be found that the energy of the range-compressed SAR data is concentrated at $t_r = (2R_0/c)$, which is independent of the azimuth velocity.

By azimuth compressing $s_{rc}(t_r, t_a)$, the focused SAR image data can be obtained as

$$s_{ac}(t_r, t_a) = \sigma_s \cdot \mathrm{sinc}\left[B_r\left(t_r - \frac{2R_0}{c}\right)\right] \cdot \mathrm{sinc}(B_a t_a) \cdot \exp\left(-\frac{j4\pi f_c R_0}{c}\right) \cdot \exp\left(-\frac{j2\pi f_c V_a^2 t_a^2}{cR_0}\right) \tag{5}$$

where B_a is the Doppler bandwidth, which is caused by the azimuth displacement of the radar.

B. Analysis

Based on the analysis of the spaceborne SAR imaging process, the ship echo signal in $s_{rc}(\cdot)$ is focused on the same range cell, which makes the ship target can be detected. In the ocean, there are few types of electromagnetic scatterers, almost only waves and a few ships. Generally, oceangoing ships, being large and equipped with significant metal components, exhibit high radar reflectivity. In contrast to sea spikes, the SAR echo signals of ships exhibit higher energy and a greater number of scattering points. Therefore, in range-compressed SAR data, the echo signals of ship and sea clutter can be distinguished. Regarding the issues mentioned in Section I, they can be solved as follows:

1) *For Sea Clutter Modeling:* According to (4) and (5), the process of converting range-compressed SAR data into the image domain is analytical. Therefore, it is feasible to derive the range-focused domain model based on the sea clutter model in the image domain.
2) *For Computational Complexity:* When SAR scans the vast ocean, there is usually only sea clutter echo. Thus, in modeling sea clutter using GΓD, computational complexity can be optimized by fine-tuning the accuracy.
3) *For Sea Spike Interference:* It is evident that the high-intensity scattering points from ships form a continuous pattern, in contrast to the isolated sea spikes. Accordingly, a fine extractor for detecting ships can be customized.

In summary, it is practical to detect ships from range-compressed SAR data. To facilitate understanding, Fig. 2 shows the working paradigm of detecting ships from range-compressed SAR data. Subsequently, the RC-GΓD model and RC-GΓD-CFAR detection method will be introduced in detail.

III. Sea Clutter Modeling in Range-Compressed SAR Data

In this section, the sea clutter model GΓD is presented at first. Then, based on GΓD, the model of range-compressed SAR sea clutter data is derived. Finally, the analytical solution of the model parameter estimation is given.

A. GΓD Model

Let x denote the amplitude of sea clutter, the probability density function (pdf) of GΓD can be expressed as [27]

$$p_{\mathrm{G\Gamma D}}(x) = \frac{|v|\kappa^\kappa}{\sigma\Gamma(\kappa)}\left(\frac{x}{\sigma}\right)^{\kappa v - 1} \exp\left\{-\kappa\left(\frac{x}{\sigma}\right)^v\right\} \quad \sigma, |v|, \kappa, x > 0 \tag{6}$$

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where σ, υ, and κ refer to the scale, power, and shape attributes, respectively. When $\upsilon = 1$, (6) is Gamma distribution; when $\kappa = 1$, (6) is Weibull distribution; when $\upsilon = 2$ and $\kappa = 1$, (6) is Rayleigh distribution; when $\upsilon = 1$ and $\kappa = 1$, (6) is exponential distribution. It is worth noting that (6) is also one of the various expressions of GΓD [32], which is also the most robust one. Due to the form of $|\upsilon|$, $\upsilon = -2$ and $\upsilon = -1$ can be regarded as the square root of inverse Gamma and inverse Gamma distribution, respectively, which is of great benefit to the development of new heterogeneous statistical models.

In mathematical statistics theory, fitting distribution is to find a pdf to describe the behavior of a set of observed data as well as possible. Therefore, directly employing existing models to fit the clutter in the range-focused domain is acceptable. However, according to signal processing theory, the distribution model of the same signal usually varies across different data domains [33], [34]. As for SAR echo signals, from (4) and (5), there are significant differences between the range-compressed domain and image domain. Hence, applying the image domain pdf model directly to range-compressed data lacks rigorous theoretical justification.

The initial fitting of a data model is often very time-consuming and laborious. Therefore, it is very meaningful to adapt mature sea clutter models to the range-focused domain. In particular, the GΓD model can fit sea clutter well and calculate detection thresholds in a closed form. These characteristics are very important for SAR real-time ship monitoring systems.

B. Proposed RC-GΓD Model

According to the second chapter of [35], when there is a function $y = f(x)$ and the pdf $p_y(y)$ is known, the pdf $p_x(x)$ can be obtained, which is expressed as

$$p_x(x) = p_y(y)\left|\frac{dy}{dx}\right| = p_y(f(x))\left|f'(x)\right|. \tag{7}$$

An important condition for (7) is that $f(x)$ is a linear transformation. Therefore, it is the key to proving that the transformation of sea clutter data from the range-focused domain to the image domain is a linear operation.

Considering only sea clutter scattering, the echo signal y_{signal} on each pixel in the SAR image is described as

$$y_{\text{signal}} = \sum_{i=1}^{N} s_{ac,i}(t_r, t_a) \tag{8}$$

where N denotes the number of echo signals. This means that the signal at each point is the energy accumulation of the surrounding N sea clutter echoes. Because each sea clutter is independent and identically distributed, the amplitude probability distribution of a single clutter also obeys the GΓD. $s_{ac}(t_r, t_a)$ contains both amplitude and phase components, where the amplitude component is the key to determining the probability density distribution of amplitude. Further, it can be found that the exponential terms of (4) and (5) are the same, which means that the phase component of the signal does not change. Therefore, only the change of the nonexponential term of (5) should be considered, as follows:

$$y_{ac} = \operatorname{sinc}\left[B_r\left(t_r - \frac{2R_0}{c}\right)\right] \cdot \operatorname{sinc}(B_a t_a). \tag{9}$$

Bringing (9) into (6), the amplitude probability density distribution function of SAR images can be obtained as follows:

$$p(y_{ac}) = \frac{|\upsilon|\kappa^{\kappa}}{\sigma\Gamma(\kappa)} \cdot \left(\frac{y_{ac}}{\sigma}\right)^{\kappa\upsilon-1} \cdot \exp\left\{-\kappa\left(\frac{y_{ac}}{\sigma}\right)^{\upsilon}\right\}, \quad \sigma, |\upsilon|, \kappa, y_{ac} > 0. \tag{10}$$

The amplitude component of range-compressed data $s_{rc}(t_r, t_a)$ as

$$x_{rc} = w_a(t_a) \cdot \operatorname{sinc}\left[B_r\left(t_r - \frac{2R_0}{c}\right)\right]. \tag{11}$$

Now, we can find that the relationship between x_{rc} and y_{ac} is as follows:

$$y_{ac} = \frac{\operatorname{sinc}(B_a t_a) \cdot x_{rc}}{w_a(t_a)} \tag{12}$$

where $w_a(t_a)$ is a rectangular window function with amplitude 1. The value of $w_a(t_a)$ in the same range cell is the same, so this term can also be ignored. Hence, (12) can be simplified as follows:

$$y_{ac} = \operatorname{sinc}(B_a t_a) \cdot x_{rc} \tag{13}$$

where B_a is the Doppler bandwidth, t_a is the azimuth slow time, both of which belong to the inherent characteristics of SAR. Obviously, (13) is a linear transformation equation. Thus, the condition for (7) to hold is met. Based on (13), it can be obtained that

$$f'(x_{rc}) = \frac{dy_{ac}}{dx_{rc}} = \operatorname{sinc}(B_a t_a) = \frac{\sin(\pi B_a t_a)}{\pi B_a t_a}. \tag{14}$$

In order to simplify the expression, let $\lambda = (1/|\operatorname{sinc}(B_a t_a)|)$, combined with (7), (10), (13), and (14), the sea clutter amplitude pdf of range-compressed data is obtained as

$$p_{rc}(x) = p_{ac}(y) \cdot \left|\frac{dy}{dx}\right| = p_{ac}(y) \cdot |\operatorname{sinc}(B_a t_a)| = \frac{|\upsilon|\kappa^{\kappa}}{\sigma\lambda\Gamma(\kappa)} \cdot \left(\frac{x}{\sigma}\right)^{\kappa\upsilon-1} \cdot \exp\left\{-\kappa\left(\frac{x}{\sigma}\right)^{\upsilon}\right\}$$
$$\sigma, |\upsilon|, \kappa, \lambda, x > 0. \tag{15}$$

For range-compressed SAR data, a sea clutter model based on GΓD is derived, which is called RC-GΓD model.

C. Parameter Estimation

Before applying a new model to CFAR, parameter estimation methods should be provided. As for RC-GΓD model, it refers to parameters υ, σ, and κ. The Mellin transform of the pdf is called the second-kind first characteristic function [36], as follows:

$$\phi(s) = \int_0^{+\infty} p(x)x^{s-1}dx. \tag{16}$$

In addition, the second-kind second characteristic function of $\phi(s)$ is its logarithmic form as

$$\varphi(s) = \log(\phi(s)). \tag{17}$$

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The second-kind cumulants of $\varphi(s)$ is written as

$$c_i = \frac{d^i}{ds^i}\varphi(s)\bigg|_{s=1}. \tag{18}$$

According to the description of [25], as long as the value of c_i is obtained, the parameters in (15) can be estimated. Then, the Mellin transform of (15) can be obtained as

$$\phi(s) = \frac{|v|\kappa^\kappa}{\sigma^{\kappa v}\lambda\Gamma(\kappa)}\int_0^{+\infty} x^{\kappa v+s-2}\exp\left\{-\kappa\left(\frac{x}{\sigma}\right)^v\right\}dx. \tag{19}$$

Given $v \neq 0$, we first consider $v > 0$. According to [27], there is an identity expressed as

$$\int_0^{+\infty} x^{\gamma-1}\exp\{-\eta x^\varrho\}dx = \frac{1}{\varrho}\eta^{-\gamma/\varrho}\Gamma\left(\frac{\gamma}{\varrho}\right) \tag{20}$$

which holds if $\gamma > 0$. When solving c_i in this topic, $s = 1$, $\kappa > 0$, and $v > 0$, so $\kappa v + s - 1 > 0$, then the condition holds. Thus, (19) can be converted to the form of (20) as follows:

$$\phi(s) = \frac{\Gamma\left(\kappa + \frac{s}{v} - \frac{1}{v}\right)}{\lambda\Gamma(\kappa)}\sigma^{s-1}\kappa^{\frac{(1-s)}{v}}. \tag{21}$$

Similarly, when $v < 0$, we can get the same form as (21). Bring (21) into (17), and then bring the result into (18). The following results can be obtained:

$$\begin{aligned} c_1 &= \frac{d^i}{ds^i}\log(\phi(s))\bigg|_{i=1,s=1} = \log(\sigma) + \frac{\Phi_0(\kappa) - \log\kappa}{v} \\ c_n &= \frac{d^i}{ds^i}\log(\phi(s))\bigg|_{i=n,s=1} = \frac{\Phi_0(n-1,\kappa)}{v^n}, \quad n = 2, 3, 4, \ldots \end{aligned} \tag{22}$$

where $\Phi_0(x)$ is the logarithmic derivative of the Gamma function, called the Digamma function. $\Phi_0(n, x)$ is the nth Polygamma function, which is also the nth derivative of the Digamma function. Since the parameter λ is independent of s, it is eliminated in the derivation calculation. Three parameters can be estimated according to three second-kind cumulant equations

$$c_1 = \ln\sigma + (\Phi_0(x) - \ln\kappa)/v \tag{23a}$$
$$c_2 = \Phi_0(1,\kappa)/v^2 \tag{23b}$$
$$c_3 = \Phi_0(2,\kappa)/v^3. \tag{23c}$$

Meanwhile, by combining (23b) and (23c), we obtain the following equation containing only the parameter κ:

$$g(\kappa) = \frac{c_2^3}{c_3^2} = \frac{{\Phi_0}^3(1,\kappa)}{{\Phi_0}^2(2,\kappa)}. \tag{24}$$

According to the description in [23], c_2 and c_3 can be estimated by the samples of the clutter. Then, the only parameter κ in (24) can be solved. Let $\{x_i\}, i \in [1, N]$ denote the sampled sea clutter dataset, where N denotes the number of samples in the data. So, the first three-order empirical log-cumulants estimation method can be calculated through [36]

$$\hat{c}_1 = \frac{1}{N}\sum_{i=1}^{N}\ln x_i \tag{25a}$$
$$\hat{c}_2 = \frac{1}{N}\sum_{i=1}^{N}\left(\ln x_i - \hat{c}_1\right)^2 \tag{25b}$$
$$\hat{c}_3 = \frac{1}{N}\sum_{i=1}^{N}\left(\ln x_i - \hat{c}_1\right)^3 \tag{25c}$$

where the sharp angle above the letter denotes the estimated second-kind cumulants. Generally, combined with (24), $\hat{\kappa}$ can be estimated directly by numerical calculation. Considering the real-time processing scene of spaceborne SAR in this article, this method has the following problems:

1) According to (24), the numerical calculation process includes solving the Polygamma function, which increases the computational complexity and time consumption.
2) The traditional numerical calculation method requires a priori initial value, which has a great influence on the calculation results. Therefore, it cannot be applied to highly automated digital signal processing systems.

Given these issues, we use an approximate estimator to calculate the value of κ. First, based on the asymptotic formula for Polygamma proposed by Li et al. [27], the approximations of $\Phi_0(1,\kappa)$ and $\Phi_0(2,\kappa)$ are as follows:

$$\Phi_0(1,\kappa) \approx \frac{1}{\kappa} + \frac{1}{2\kappa^2} \tag{26a}$$
$$\Phi_0(2,\kappa) \approx -\frac{1}{\kappa^2} - \frac{1}{\kappa^3}. \tag{26b}$$

This approach can further improve the efficiency of parameter estimation while ensuring calculation accuracy. Bringing (26) into (24), we can have

$$g(\kappa) = \frac{\Phi_0^3(1,\kappa)}{\Phi_0^2(2,\kappa)} \approx \frac{\left(\frac{1}{\kappa} + \frac{1}{2\kappa^2}\right)^3}{\left(\frac{1}{\kappa^2} + \frac{1}{\kappa^3}\right)^2} = \frac{\left(\kappa + \frac{1}{2}\right)^3}{(\kappa+1)^2}. \tag{27}$$

Next, combining (24) and (27), the following equation can be obtained:

$$\kappa^3 + \left(\frac{3}{2} - \frac{c_2^3}{c_3^2}\right)\kappa^2 + \left(\frac{3}{4} - 2\frac{c_2^3}{c_3^2}\right)\kappa + \frac{1}{8} - \frac{c_2^3}{c_3^2} = 0. \tag{28}$$

It can be considered an ordinary one-variable cubic equation. Since κ is a positive value, according to Cardano's formula, it can be obtained by [27]

$$\kappa = \frac{2c_2^3 - 3c_3^2}{6c_3^2} + \sqrt[3]{-\frac{q}{2} + \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} \tag{29}$$

$$p = \frac{3a_0a_2 - a_1^2}{3a_0^2} \tag{30}$$

$$q = \frac{2a_1^3 - 9a_0a_1a_2 + 27a_0^2a_3}{27a_0^3} \tag{31}$$

$$a_0 = 8c_3^2, \quad a_1 = 4\left(3c_3^2 - 2c_2^3\right)$$
$$a_2 = 2\left(3c_3^2 - 8c_2^3\right), \quad a_3 = c_3^2 - c_2^3. \tag{32}$$

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It should be emphasized that condition $3c_3^2 \leq 8c_2^3$ is necessary to ensure the existence of positive real solutions.

Finally, after using (25) to estimate $\hat{c}_2$ and $\hat{c}_3$, $\hat{\kappa}$ can be directly calculated. As for $\hat{v}$ and $\hat{\sigma}$, they can be calculated by (23a) and (23b) as [28]

$$\hat{v} = \text{sign}\left(-\hat{c}_3\right) \cdot \sqrt{\frac{1}{\hat{c}_2}\Phi_0\left(1, \hat{\kappa}\right)} \tag{33a}$$

$$\hat{\sigma} = \exp\left\{\hat{c}_1 - \frac{\Phi_0\left(\hat{\kappa}\right) - \ln\left(\hat{\kappa}\right)}{\hat{v}}\right\}. \tag{33b}$$

Algorithm 1 summarizes the whole parameter estimation method.

Algorithm 1 Parameter Estimation of the pdf of RC-GΓD

1: **Input:** range-compressed SAR data samples $\{x_i\}, i \in [1, N]$, azimuth linear modulation frequency K_a, signal pulsewidth T_p.
2: Compute $\lambda = 1/|sinc(B_a t_a)|$;
3: Compute $\hat{c}_1$, $\hat{c}_2$, and $\hat{c}_3$ via (25a), (25b), and (25c), respectively;
4: Compute p, q, a_0, a_1, a_2, and a_3 via (30), (31), and (32);
5: Compute $\hat{\kappa}$ via (29);
6: Compute $\hat{v}$ and $\hat{\sigma}$ via (33a) and (33b).
7: **Output:** pdf of RC-GΓD $p_{rc}(x)$.

IV. CFAR-Based Ship Detection in Range-Compressed SAR Data

In this section, the RC-GΓD-CFAR method is presented. First, based on the distribution characteristics of the sea clutter in the SAR range-focused domain, a customized data sampling method is demonstrated. Then, a fast adaptive threshold calculation method adapted to RC-GΓD is given. Finally, due to the existence of clutter false alarm points in the binary results, a fine ship extractor is proposed.

A. Data Sampling

Data sampling determines the accuracy of model parameter estimation. The smaller the proportion of nonclutter data in the sampling data, the more accurate the modeling. The target energy in the image domain is focused, making it easy to sample clutter data. However, there are problems in using classical data sampling methods for range-compressed data.

1) *Local Sampling:* Sliding windows, protective windows, and background windows are used. The clutter distribution in the azimuth and range directions is not uniform, so the data sampled by the ordinary window is not representative.
2) *Global Sampling:* Due to the large scanning area of spaceborne SAR, this method brings a large computational burden. At the same time, more nonclutter data are sampled.

Therefore, the classical CFAR data sampling method cannot be directly applied in this study. Based on the data characteristics in the range-focused domain, a nonlocal sampling method with dense range and sparse azimuth is customized. The detailed steps are as follows:

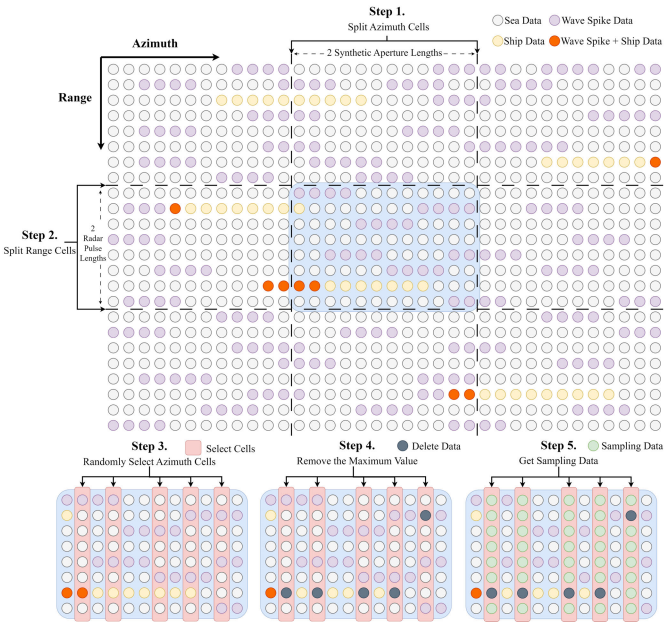


Fig. 3. Schematic of nonlocal sampling of 2-D range-compressed SAR data, where the blue box represents the target area. Purple, yellow, and orange circles represent clutter, ship, and mixed energy data, respectively.

Step 1: The range-compressed data are split along the azimuth direction, with an interval of two synthetic aperture lengths.

Step 2: The range-compressed data are split along the range direction, with an interval of two radar pulse lengths.

Step 3: A patch is selected as the target area, and N azimuth cells are sparsely selected from it.

Step 4: The maximum value in each azimuth cell is removed.

Step 5: The remaining data of the target area are used as a sample set of clutter.

For the convenience of understanding, the range-compressed data sampling method is shown in Fig. 3. The proposed nonlocal method for range-compressed data sampling has the following two main advantages:

1) *Comprehensive Characteristics:* All the range cells are collected to ensure that the sea clutter energy is not missed. The method of sparse sampling azimuth cells reduces redundant data while enriching characteristics.
2) *Strong Effectiveness:* By removing the maximum value, the interference caused by the ship signal or abnormal signal can be reduced. Due to the few high intensity points in sea clutter, even if removed in Step 4, it does not affect the goodness-of-fit of the model.

After data sampling, the model parameters can be estimated according to Algorithm 1.

B. Adaptive Threshold Calculation

For CFAR, calculating the detection threshold is a key step. By comparing the pixel under test (PUT) with threshold T, the binary result of the preliminary detection can be obtained. Generally, the threshold is estimated through iterative methods,

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resulting in a heavy computational burden, making it unsuitable for spaceborne SAR systems. One advantage of the RC-GΓD model is that its adaptive threshold can be analytically solved. The cumulative distribution function $F(x)$ of the sea clutter pdf in the range-focused domain can be expressed as

$$F(x) = \int_0^x p(t)dt. \tag{34}$$

The probability of false alarm (PFA) is as follows:

$$P_{\text{fa}} = \int_T^{\infty} p(x)dx = 1 - F(T). \tag{35}$$

Bring (15) into (34), and let $y = \kappa((x/\sigma))^v$, the cumulative distribution function can be described as

$$F(x) = \begin{cases} \dfrac{1}{\lambda\Gamma(\kappa)} \displaystyle\int_0^{\kappa\left(\frac{x}{\sigma}\right)^v} y^{\kappa-1}e^{-y}dy, & v > 0 \\ \dfrac{1}{\lambda\Gamma(\kappa)} \displaystyle\int_{\kappa\left(\frac{x}{\sigma}\right)^v}^{\infty} y^{\kappa-1}e^{-y}dy, & v < 0. \end{cases} \tag{36}$$

Bringing (36) into (35) is a classical numerical method for calculating the threshold T. In this article, based on the incomplete Gamma definition, a method for calculating the threshold T without iteration is given. The incomplete Gamma equation is defined as

$$Q(x, a) = \frac{1}{\Gamma(a)} \int_0^x t^{a-1} \exp(-t)dt, \quad a, x > 0. \tag{37}$$

Combining (36) and (37), another form of clutter cumulative distribution function in the range-focused domain can be obtained as

$$F(x) = \begin{cases} \dfrac{Q\left(\kappa\left(\frac{x}{\sigma}\right)^v, \kappa\right)}{\lambda}, & v > 0 \\ 1 - \dfrac{Q\left(\kappa\left(\frac{x}{\sigma}\right)^v, \kappa\right)}{\lambda}, & v < 0. \end{cases} \tag{38}$$

Based on (38) and (35), the analytical expression of the adaptive threshold T is as follows:

$$T = \begin{cases} \sigma\left\{\dfrac{1}{\kappa} Q_{\text{Inv}}\left(\lambda - \lambda P_{\text{fa}}, \kappa\right)\right\}^{\frac{1}{v}}, & v > 0 \\ \sigma\left\{\dfrac{1}{\kappa} Q_{\text{Inv}}\left(\lambda P_{\text{fa}}, \kappa\right)\right\}^{\frac{1}{v}}, & v < 0 \end{cases} \tag{39}$$

where P_{fa} is usually set to a constant value less than 0.01 [37], and $Q_{\text{Inv}}(\cdot)$ denotes the inverse incomplete Gamma equation. For the calculation of $Q_{\text{Inv}}(\cdot)$, please refer to the asymptotic expression method described in [38].

C. Ship Data Extraction

Comparing the PUT with the threshold can remove most of the clutter pixels. However, due to the unfocused energy in the azimuth direction, the ship and clutter pixels are easily confused. Hence, it is necessary to further eliminate false alarm pixels. During SAR scanning, ship displacement is limited. Therefore, the scattering echo of the ship is stable and continuous. Due to the endless changes in the sea surface, the scattering echo of the spike is unstable. It can be noted that different from ships, wave spikes are high-intensity isolated points. Therefore, a fine ship extractor is customized for the preliminary detection results.

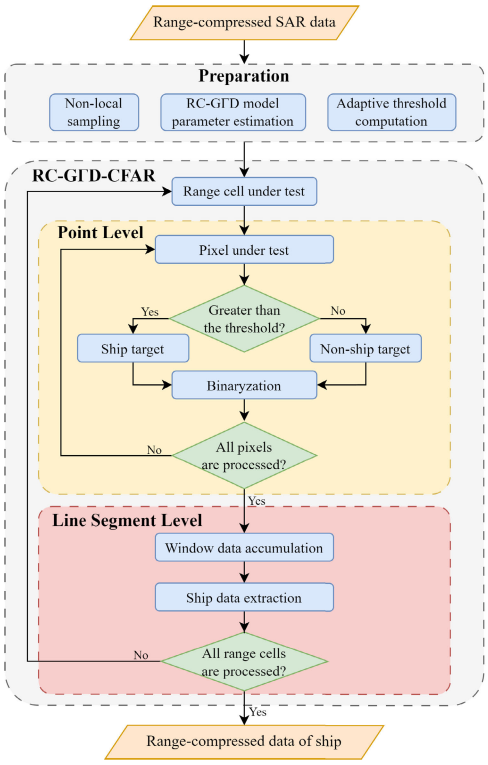


Fig. 4. Flowchart of the RC-GΓD-CFAR method.

Up to this point, the complete content of the proposed RC-GΓD-CFAR method is shown in Fig. 4. In range-focused domain, comparing each pixel with the detection threshold can be regarded as point-level processing. Then, the binary results are filtered by a 1-D sliding window with two synthetic aperture lengths. When there is a high accumulation within the window, the azimuth data segment is preserved; otherwise, it is discarded. This can be regarded as a line segment-level processing.

By compressing the extracted azimuth data segments, SAR ship images can be obtained. Surprisingly, the noise is reduced and ship targets very clear. We believe that this is because the proportion of ship targets in the imaging data increases and the interference decreases.

V. Experimental Results and Analysis

In this section, the experimental results are presented and analyzed. First, the test dataset and experimental details are introduced. Next, the performance of the RC-GΓD model in fitting sea clutter in the SAR range-focused domain is investigated. Finally, the effectiveness of the RC-GΓD-CFAR method is also demonstrated.

A. Dataset Description and Experimental Details

In this section, all experiments are conducted on four sets of real raw SAR data. For the convenience of comparison, the key information of each data is listed in Table I. These real

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TABLE I
KEY INFORMATION OF FOUR REAL RAW SAR DATA

Test Data	Sensor	Region	Date	Range Sampling Frequency (MHz)	Polarization Mode	Acquisition Mode	Size ($Range \times Azimuth$)	Pulse Repetition Frequency (Hz)	Wavelength (m)
Data-1	ERS-2	Mediterranean West Coast	2004/09/23	18.962	VV	Strimap	4912×27600	1659.904	0.057
Data-2	ERS-2	Alboran Sea	1994/09/23	18.962	VV	Strimap	4912×27600	1659.904	0.057
Data-3	Sentinel-1	East China Sea	2021/07/26	64.345	VV	TOPSAR	23886×1409	1717.129	0.056
Data-4	Sentinel-1	Cape of Good Hope	2023/09/14	64.345	VH	TOPSAR	19514×1410	1717.129	0.056

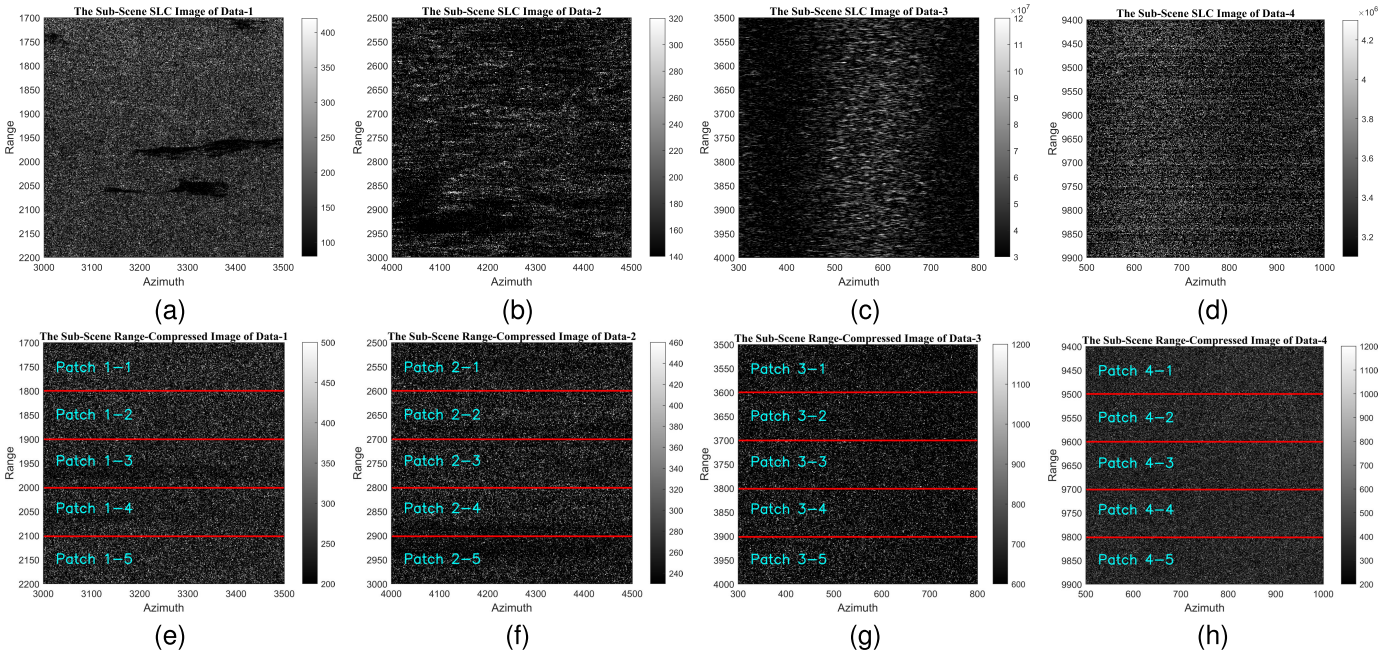


Fig. 5. Subscenes of real SAR data used in the experiment. The subscene SLC images of Data-1, Data-2, Data-3, and Data-4 are (a)–(d), respectively, and the corresponding range-compressed data are (e)–(h), respectively.

raw SAR data, which are provided free of charge by the ASF website, are taken from different sensors and regions.

Here, we briefly explain how to work on these raw data. For ERS-2, according to the description in [39], we extract the ERS system parameters and obtain valid raw data. Refer to the processing of SAR data in [31], the ERS-2 range-compressed data for the experiment is ready. As for Sentinel-1, its raw data processing method is complex and there are no public projects. At first, we mainly refer to [40] and [41] to decode, check, and construct the Sentinel-1 SAR space packet, and extract the radar parameters. Then, the raw SAR data are analyzed based on [42], including I/Q data decoupling, bias removal, gain balance correction, and nonorthogonality correction. It is worth mentioning that the acquisition mode of Sentinel-1 is terrain observation by progressive scans SAR (TOPSAR). For each burst, the beam is electronically steered from backward to forward in the azimuth direction. Then, the range reference function is extracted from each burst and used it for range compressing. Finally, the Sentinel-1 range-compressed data for the experiment is also ready.

In brief, the imaging process of each sensor involves many signal processing algorithms. This is not the focus of this article. To facilitate understanding, related projects designed and developed for this study are uploaded in (https://github.com/GandalfGondor/RawSARDataProcessor).

B. Validation of RC-GΓD Model

In this subsection, the performance of the proposed RC-GΓD model in fitting sea clutter in range-focused domain is analyzed.

1) Test Dataset: Because the scene size of the real raw SAR data is too large, four subscenes without ships are extracted from Data 1–4 for experiments, as shown in Fig. 5. Fig. 5(a)–(d) are SLC images with bright and dark textures or patches, which are formed by the scattering of peaks and valleys of sea clutter. In addition, the sea conditions of Fig. 5(a) and (c) are relatively stable, with large homogeneous areas. Fig. 5(b) and (d) are relatively turbulent, with a majority of heterogeneous areas. Fig. 5(e)–(h) are range-compressed data, which are chaotic and cannot identify semantic information from them. The clutter fitting experiment of RC-GΓD is conducted on range-compressed data.

The sea states of different local sea regions vary in large scenes, and in the range-compressed domain, the scattered energy of sea clutter echoes is dispersed within the range cell. To verify the fitting of (19) under different sea states, the sea clutter data should also be fully sampled. Then, we divide the test subscenes along the range direction. Specifically, in Fig. 5, subscenes are divided into patches with 100×500 (range $\times$ azimuth) pixels by red lines, numbered patches $n-1, \ldots, n-5$ in order.

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TABLE II
PARAMETER ESTIMATION RESULTS

Patch No.	Log-normal (log location, log scale)	Weibull (shape, scale)	K+Noise (shape)	K+Rayleigh (shape)	GΓD (scale, power, shape)	3MD (a_1, a_2, a_3) (c_1, c_2, c_3)	RC-GΓD (scale, power, shape)
Patch 1-1	(5.125, 0.644)	(236.861, 2.039)	5.563	2.694	(0.953, 0.527, 43.276)	(0.600, 1.371, 3.017) (0.739, 0.673, 0.533)	(0.943, 0.577, 45.337)
Patch 2-1	(4.985, 0.648)	(213.173, 1.917)	4.572	2.078	(0.721, 0.601, 37.407)	(0.525, 0.977, 2.213) (0.791, 0.715, 0.243)	(0.767, 0.599, 40.525)
Patch 3-1	(4.631, 0.645)	(138.061, 2.095)	3.811	1.924	(0.874, 0.713, 26.117)	(0.441, 0.732, 1.618) (0.371, 0.266, 0.113)	(0.889, 0.761, 29.332)
Patch 4-1	(4.812, 0.643)	(150.753, 1.933)	0.863	0.431	(0.905, 0.482, 34.042)	(0.491, 0.092, 0.014) (0.471, 0.365, 0.281)	(0.927, 0.501, 35.296)

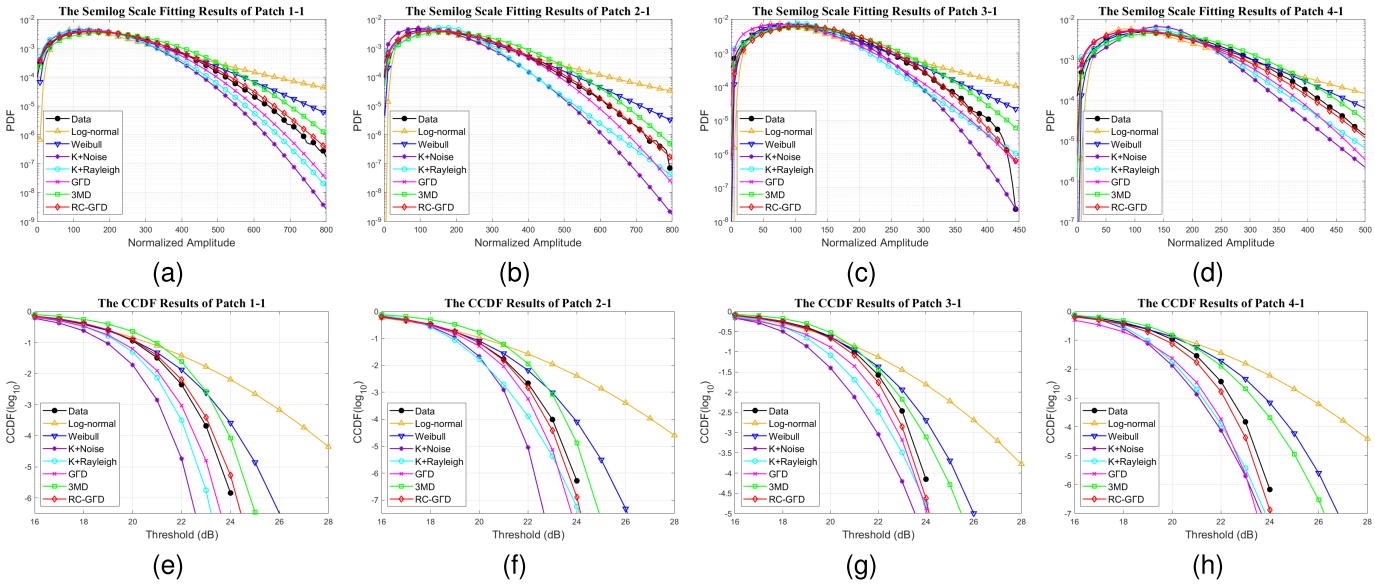


Fig. 6. Amplitude distributions for four test datasets. (a), (b), (c), and (d) Fitting PDF results of patches 1-1, 2-1, 3-1, and 4-1 on semilog scale, respectively. (e), (f), (g), and (h) CCDF results of patches 1-1, 2-1, 3-1, and 4-1, respectively.

2) Fitting Results: According to Section III-C, the parameters of the RC-GΓD model, such as scale, power, and shape are estimated, respectively. To highlight the performance of the proposed method, some classical models, such as Log-normal, Weibull, K + Noise, K + Rayleigh, GΓD, and 3MD are compared by experiments. For each subscene, only one patch test result is displayed. Table II lists the parameters of all models estimated by Patches 1-1, 2-1, 3-1, and 4-1, respectively. By visualizing the complementary cumulative distribution function (CCDF) and semilogarithmic fitting results, Fig. 6 shows the goodness-of-fit of all models.

3) Quantitative Evaluation: It is also important to quantitatively evaluate the goodness-of-fit accuracy of each model. Symmetric Kullback–Leibler (KL) distance is proposed [43], which can be used to analyze the global similarity between the estimated pdf and the distribution of the real data. When the value of symmetrical KL is large, it is considered that the estimated pdf has a large difference from the data distribution histogram, and the goodness-of-fit is poor. On the contrary, the goodness-of-fit is good.

Let Q, H, and G denote the test data, the received data, and the estimated data, respectively. $p_H(x)$ and $p_G(x)$ denote the statistical distribution and estimated probability distribution of the received data, respectively. Then, the symmetric KL distance is calculated as follows:

$$D_{\mathrm{KL}}(H, G) = \sum_{x \in Q} p_H(x) \log \frac{p_H(x)}{p_G(x)} + \sum_{x \in Q} p_G(x) \log \frac{p_G(x)}{p_H(x)}. \tag{40}$$

The symmetric KL mainly focuses on the body fitting effect of the data, and the evaluation of the tail data is insufficient. The tailed KL (TKL) index is complementary to it, focusing on evaluating the fitting effect of the tail data. Furthermore, the purpose of estimating the probability density of sea clutter is ship detection. The amplitude of the ship is mostly distributed at the tail, so the fitting accuracy of pdf at the tail is more important. The difference between TKL and KL is the test dataset Q, which selects the data with the amplitude in the top 10%.

According to [22], threshold error is also an important metric. It is the difference between the fitting result and the real data at a fixed CCDF value (usually 10^{-4}). The smaller the difference, the better the goodness-of-fit. It is worth noting that CCDF plays the role of PFA in clutter fitting and determines the CFAR threshold.

So far, the evaluation indexes used in this experiment are all introduced. Table III shows the mean numerical evaluation results of different distribution functions on each subscene.

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TABLE III
QUANTITATIVE COMPARISON OF MEAN FITTING PERFORMANCE BETWEEN DIFFERENT DISTRIBUTIONS

PDF Model	Mean KL Value				Mean TKL Value				Mean Threshold Error Value (dB)			
	Data-1	Data-2	Data-3	Data-4	Data-1	Data-2	Data-3	Data-4	Data-1	Data-2	Data-3	Data-4
Log-normal	0.1278	0.2550	0.1261	0.2815	0.4142	0.4186	0.1578	0.3665	-4.31	-4.06	-4.51	-4.19
Weibull	0.0740	0.1251	0.0619	0.1473	0.1507	0.2686	0.1121	0.3111	-1.21	-0.88	-1.39	-1.63
K+Noise	0.0364	0.0945	0.0551	0.1324	0.0975	0.1279	0.0907	0.2489	1.48	0.15	1.24	1.29
K+Rayleigh	0.0312	0.0826	0.0385	0.0987	0.0695	0.0742	0.0722	0.1206	0.88	0.84	0.47	0.99
GΓD	0.0293	0.0723	0.0359	0.0772	0.0581	0.0614	0.0757	0.1148	0.45	0.62	0.37	0.87
3MD	0.0253	0.0589	0.0281	0.0335	0.0494	0.0579	0.0657	0.0979	0.26	-0.50	-0.88	-1.10
RC-GΓD	**0.0136**	**0.0185**	**0.0098**	**0.0131**	**0.0174**	**0.0162**	**0.0326**	**0.0410**	**-0.19**	**0.23**	**0.58**	**0.39**

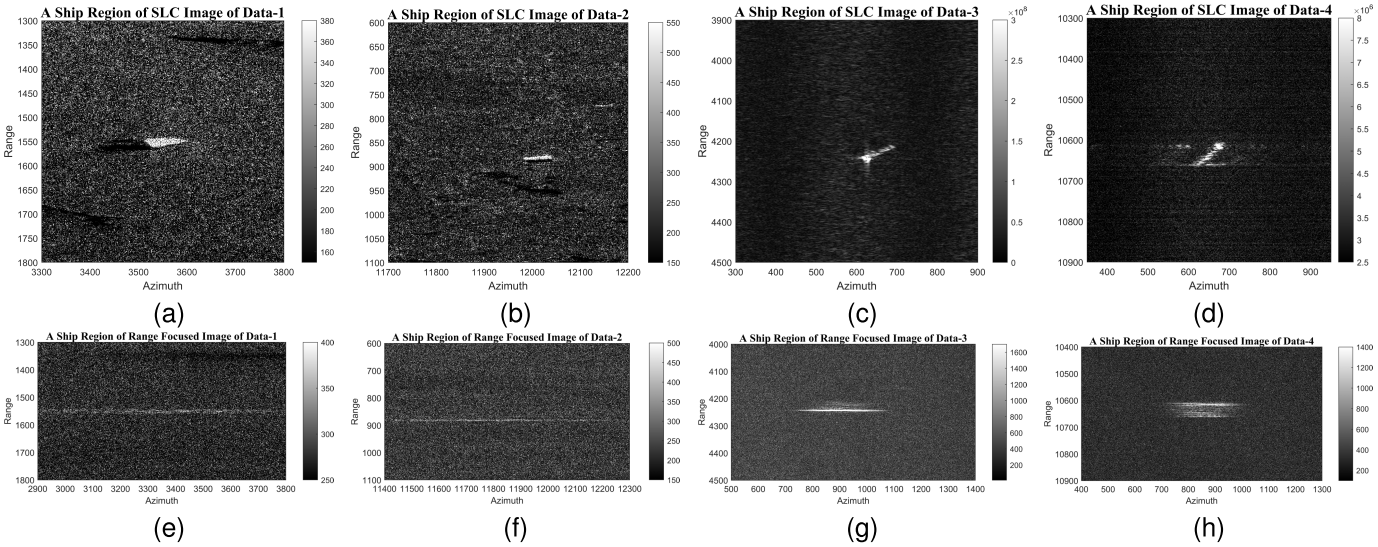


Fig. 7. SLC and range-focused images of ship regions. (a)–(d) are the SLC images taken from Data-1, Data-2, Data3, and Data-4, respectively. (e)–(h) are the range-focused images taken from Data-1, Data-2, Data3, and Data-4, respectively.

4) Analysis: As can be observed from Table III and Fig. 6, several conclusion can be drawn as follows:

1) *For ERS-2 Data:* The sea state of the Data-1 is relatively stable, with low wind speed and few peaks, while the Data-2 is more intense. Observing Fig. 6(a) and (b), Log-normal, Weibull, and K + Noise distributions do not match the data well. Although K + Rayleigh, GΓD, and 3MD are more effective, both of them have a mismatch in modeling the distribution by under and over estimating the tail, respectively. The estimated RC-GΓD is very consistent with the data. From Fig. 6(e) and (f), it can be seen that the threshold error of RC-GΓD is the smallest, which is ±0.2 dB. From Table III, it can be seen that GΓD, 3MD, and RC-GΓD all have lower KL and threshold error values. As for TKL, the value of RC-GΓD is smaller, which can be considered more suitable for fitting tail data. Hence, the RC-GΓD can better fit the range-compressed data obtained by the ERS-2 sensor.
2) *For Sentinel-1 Data:* The sea state of Data-3 is relatively calm. The Data-4 is taken from the Cape of Good Hope, with large wind speeds, many wave spikes, and complex sea clutter scattering. By observing Fig. 6(c) and (d), Log-normal, Weibull, and K + Noise still do not match the data. Although K + Rayleigh, GΓD, and 3MD fit the body of the distribution normally, there is a large deviation in the high-amplitude data part. From Fig. 6(g) and (h), it can be clearly seen that there is a large mismatch in the tail of other distribution models. RC-GΓD has a good fit until a CCDF value of 1×10^{-4}, which means it matches the range-compressed clutter data. From Table III, the above conclusions can also be well proved. It is worth noting that other models have low KL values in Data-3, but relatively high KL values in Data-4. However, the fitting performance of RC-GΓD is relatively stable. It can be concluded that the RC-GΓD can better fit the range-compressed Sentinel-1 data and has stronger adaptability.

In summary, both visual and quantitative experimental results prove that the proposed RC-GΓD is very suitable for fitting the sea clutter distribution of range-compressed SAR data. Whether it is the difference in SAR signal properties, which is caused by different sensors, or the difference in clutter amplitude, which is caused by different sea conditions, the RC-GΓD proposed in this article can achieve better fitting results.

C. Validation of RC-GΓD-CFAR Method

After modeling the sea clutter, this subsection will verify the effectiveness of the proposed RC-GΓD-CFAR method.

1) Preparation: This subsection focuses on the performance of sensing ship targets from range-compressed SAR data. Here, four regions containing ships are extracted from Data 1–4, respectively, and their SLC and range-compressed data are provided, as shown in Fig. 7. From Fig. 7(a)–(d),

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TABLE IV
LSNR (dB) OF EACH TEST SUBSCENE

Test Dataset	SLC Image	Range Focused Image
Data-1	1.601	-4.037
Data-2	2.632	-3.736
Data-3	6.608	2.993
Data-4	4.068	1.382

the position of the ship is easily recognized by the human eye, and there are waves next to the ship. Observing Fig. 7(e)–(h), the energy of the ship is focused in the range cell and very scattered in the azimuth cell. It can be seen that the ship is almost covered by sea clutter energy, making it difficult to locate.

From a different angle, the RC-GΓD-CFAR aims to achieve weak target detection in range-compressed SAR data. Therefore, in order to highlight the detection ability of the method, it is necessary to compare the signal-to-noise ratio (SNR) of ships in SLC and range-focused images. According to [44], local signal-to-noise ratio (LSNR) is used to describe the degree of enhancement of the target and the inhibition of the background, which can be expressed as follows:

$$\mathrm{LSNR} = 10\lg \frac{T_{\mathrm{mean}} - B_{\mathrm{mean}}}{\sigma_B} \tag{41}$$

where T_{mean} is the mean value of the ship region, B_{mean} is the mean value of the background region, and the σ_B is the standard deviation of the background region. In general, the background region is three times larger than the target region.

For each test dataset, Table IV shows the LSNR results. Range-compressed data has a lower SNR than SLC, so target detection is more challenging. The SNR values of ERS datasets are negative, which means that the target energy is hidden in the background signal. Hence, fitting sea clutter is very important for detecting ships in range-compressed data. Subsequently, the effectiveness of the RC-GΓD-CFAR would be verified on each test dataset.

Referring to Fig. 2, the first step is data sampling and parameter estimation to model local sea clutter. Then, according to Fig. 6, considering that the larger PFA is meaningless for target detection, the PFA is set to 0.001. Next, the adaptive threshold is calculated based on (39). In the following, the detection method based on RC-GΓD-CFAR can be validated.

2) Experiments on ERS-2 Data: First, the proposed RC-GΓD-CFAR is verified on the Data-1. Then, each value of the range-compressed data is compared with the threshold, and the binary result is shown in Fig. 8(a). Because the sea spike energy is scattered in the azimuth direction, many pixels higher than the threshold can be seen. To better observe the ship region, Fig. 8(a) marked the area corresponding to Fig. 7(e) with a green box and enlarged it in Fig. 8(b). The area where the ship is located is marked with a yellow box around the 1550 range cell. Obviously, there is more data larger than the threshold in this area, which is consistent with the range cell of the ship shown in Fig. 7(e).

Fig. 8(c) shows the sum of Fig. 8(b) along the azimuth direction, which proves that there are a lot of high-intensity data near the 1550 range cell in this area. This verifies the

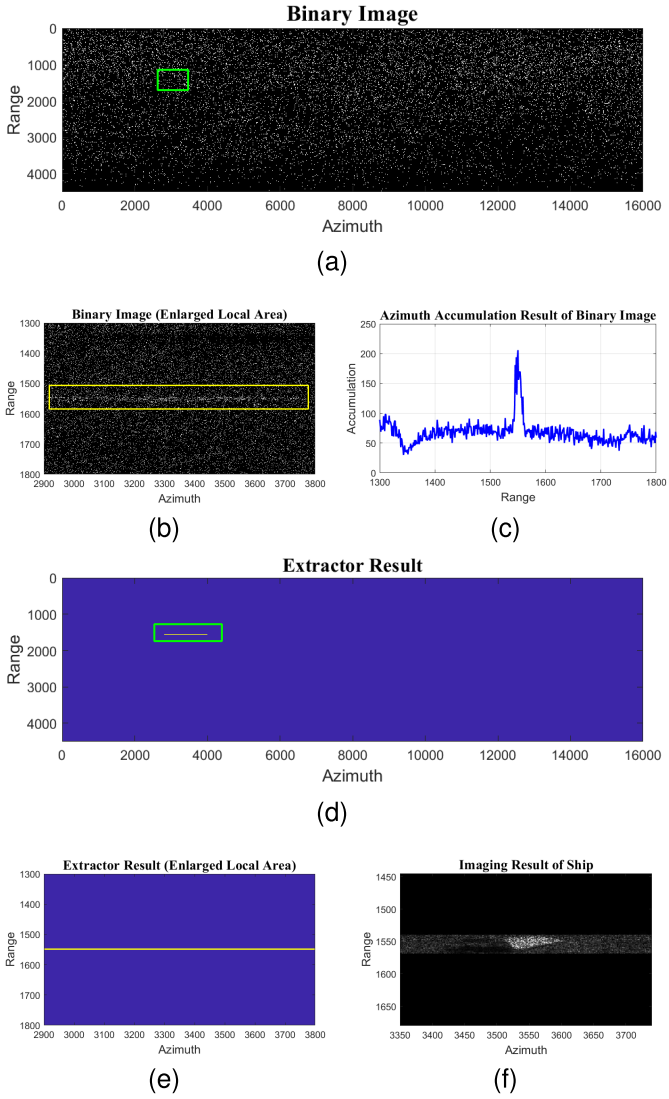


Fig. 8. Results of using the RC-GΓD-CFAR to detect ships in the Data-1. (a) Binary image. (b) Enlarged binary image. (c) Accumulative result. (d) Extractor result. (e) Enlarged extractor result. (f) Imaging result of ship.

effectiveness of the proposed RC-GΓD model. Next, according to Section IV-C, the false alarm pixels in Fig. 8(a) are eliminated as Fig. 8(d). Only one azimuth line segment is retained near 1550, which is the extracted ship target. Fig. 8(e) is the enlarged image of the detected target area. We carefully review the Data-1 SLC image and find that there is only one ship in this area, which proves that the proposed method successfully detects the ship from the range-focused domain without false alarms. Finally, the data with range between 1540 and 1570 and azimuth between 2900 and 3800 are azimuth compressed, as shown in Fig. 8(f). Compared with Fig. 7(a), this imaging result is clearer. We believe that it is because the proportion of clutter data in this area is smaller than that of the ship, which reduces the interference in imaging.

In order to verify the effectiveness of this method under different sea conditions, the same experiment is conducted on Data-2. Fig. 9(a) is the binary result image, where the green box marked area is shown in Fig. 9(b). Fig. 9(d) is the result of further extracting ships from the binary image,

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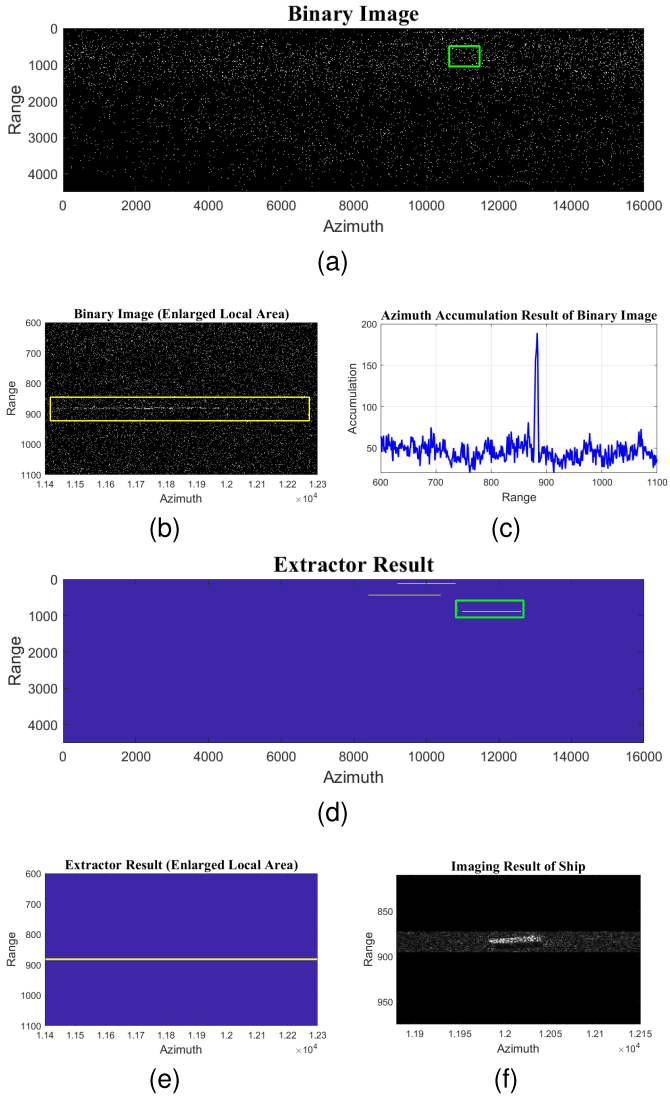


Fig. 9. Results of using the RC-GΓD-CFAR to detect ships in the Data-2. (a) Binary image. (b) Enlarged binary image. (c) Accumulative result. (d) Extractor result. (e) Enlarged extractor result. (f) Imaging result of ship.

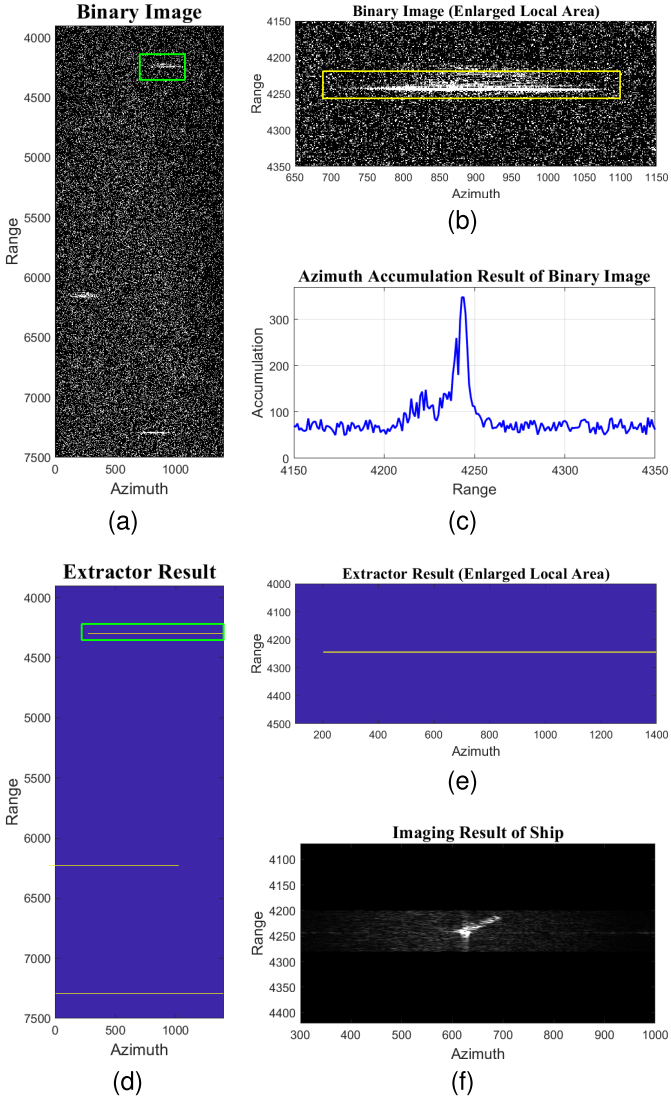


Fig. 10. Results of using the RC-GΓD-CFAR to detect ships in the Data-3. (a) Binary image. (b) Enlarged binary image. (c) Accumulative result. (d) Extractor result. (e) Enlarged extractor result. (f) Imaging result of ship.

in which multiple targets are detected. After a review of the SLC image of Data-2, the location and number of ships are correct. Fig. 9(e) is an enlarged image of one of the detection targets, which corresponds to the data shown in Fig. 9(b). The data containing the ship information is azimuth compressed, and the imaging result is shown in Fig. 9(f).

Briefly, regardless of the calm or rough sea surface sensed by ERS-2, the proposed RC-GΓG-CFAR can effectively detect ship targets in range-compressed data.

3) Experiments on Sentinel-1 Data: This part verifies the proposed RC-GΓD-CFAR method on the data collected by Sentinel-1. Following the same steps as the previous subsection, we conduct an experiment on Data-3. The binary result after threshold detection is shown in Fig. 10(a), with the green box enlarged as shown in Fig. 10(b). Fig. 10(c) is the accumulation of Fig. 10(b) along the azimuth direction. There are many pixels higher than the detection threshold at 4250. This is the same as the position of the ship shown in Fig. 7(c), which proves that RC-GΓD is also applicable to the Sentinel-1. Next, three ship targets are extracted from the binary image as shown in Fig. 10(d). In fact, it can be seen from Fig. 10(a) that there are indeed three point-intensive regions. Then, the target at the range 4240 is magnified as shown in Fig. 10(e). Finally, the data with range between 4210 and 4270 and azimuth between 300 and 1000 are compressed, and the imaging result is shown in Fig. 10(f).

To verify the effectiveness of the RC-GΓD-CFAR in complex sea-state data collected by Sentinel-1, the same experiments are conducted on Data-4. The binary result is shown in Fig. 11(a), in which the green box is enlarged as shown in Fig. 11(b). Next, the false alarm pixels in the binary results are removed, as shown in Fig. 11(d). From Fig. 11(d), two target data blocks can be seen, which is consistent with our observation of SLC images. Fig. 11(e) shows an enlarged image of the target around 10 650 in range direction, corresponding to Fig. 7(d). Finally, by azimuth compressing the range-compressed SAR data with range between 10 600 and 10 680 and azimuth between 300 and 1000, the imaging result of the ship is shown in Fig. 11(f).

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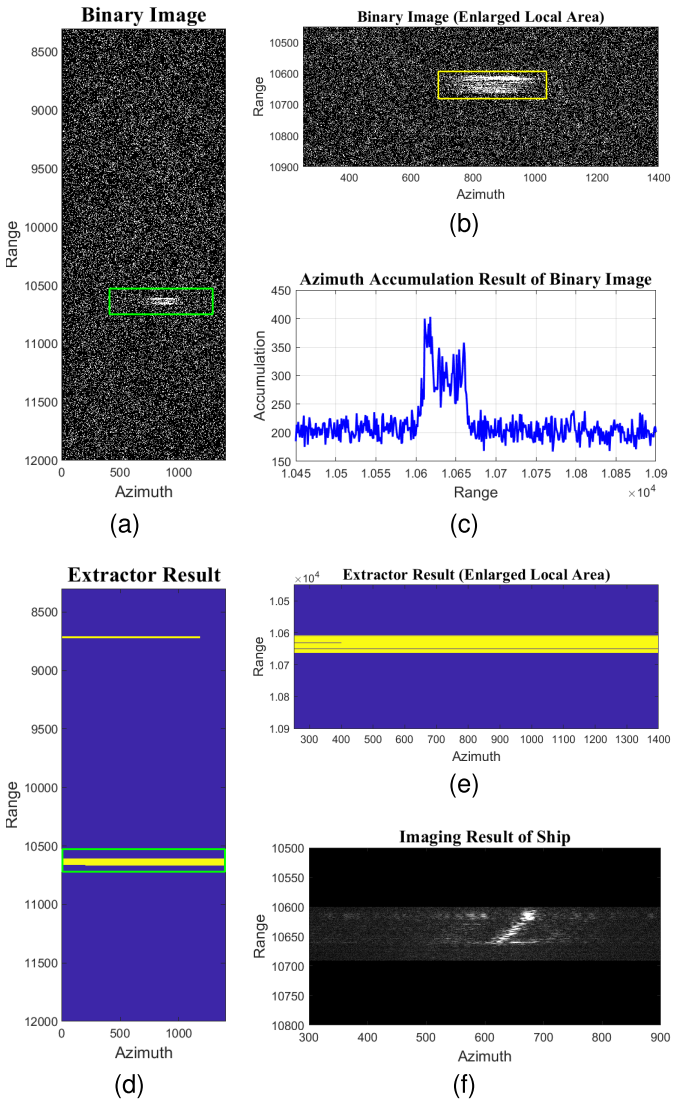


Fig. 11. Results of using the RC-GΓD-CFAR to detect ships in the Data-4. (a) Binary image. (b) Enlarged binary image. (c) Accumulative result. (d) Extractor result. (e) Enlarged extractor result. (f) Imaging result of ship.

These experiments indicate that RC-GΓD-CFAR can effectively detect ship targets in range-compressed Sentinel-1 data.

VI. Discussion

A. About Peak Detection Method

In the field of target detection, peak detection is a common method. Undoubtedly, in some simple cases, this method is effective. However, unlike image pixel values ranging from 0 to 255, the range of range-compressed data values varies greatly. The maximum value of the range-compressed data of ERS-2 is 1×10^4. Some range cells with multiple discretely distributed sea spikes have high cumulative values, which can easily be mistaken for ship targets. For Fig. 7(e), it is easy to locate the ship area through human eyes, but peak detection cannot be used to determine whether there is a ship in the range cell.

Here, a simple experiment is conducted. For the data of Data-1 between 800 and 1800 in the range direction,

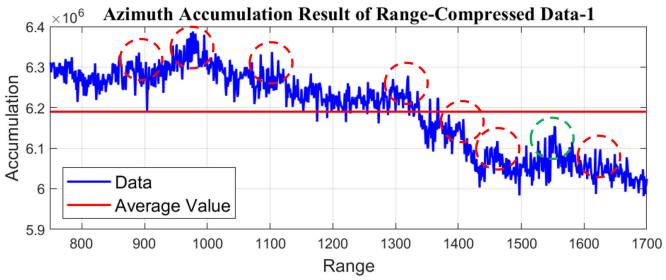


Fig. 12. Accumulated results of range-compressed data along the azimuth direction.

the average value of the total data and the cumulative value of each range cell are calculated. As shown in Fig. 12, there is a local peak at 1550 in range direction, which is the position of the ship. However, the peak value is lower than the average value. Even if the range is further reduced, the peak at 1550 is lower than the peak at 1420. The false alarm peaks, marked with red dotted circles, are difficult to eliminate. In summary, the performance of peak detection methods in range-compressed data is not robust.

B. Necessity of CFAR

Compared with the peak detection method, CFAR is more suitable for detecting targets in range-compressed SAR data.

1) *Adaptability:* The adaptive detection threshold is automatically adjusted based on different areas.
2) *Anti-Interference:* Modeling clutter in the range-focused domain to strengthen anti-interference capability.
3) *Multiscale Adaptation:* CFAR can detect targets of different scales, especially suitable for range-focused domain targets with large aspect ratio.

Therefore, CFAR can suppress most of the clutter interference pixels at the point level. Moreover, since the range-compressed data are converted to a binary image, the high-intensity clutter interference is limited. The continuity of ship scattering energy is important information that has not been utilized previously. Thus, designing a postprocessing method at the line segment level is highly meaningful. The extractor is conducted on binary data, greatly reducing the false alarm interference caused by clutter. This is a simple sequential processing module that enhances detection capabilities at a very low computational cost.

In summary, the RC-GΓG-CFAR is a variant method developed based on CFAR.

C. Application Prospect of RC-GΓG-CFAR

1) Universality: For spaceborne SAR monitoring ships, we propose a method to detect ships from range-compressed SAR data. Considering the scattered energy and the interference of sea clutter, the clutter model and ship characteristics in the SAR range-focused domain are deeply studied. Experiments are conducted on four real SAR data, and the goodness-of-fit of the designed sea clutter model is analyzed. Meanwhile, the high accuracy of the ship extractor is also proven. This method has low computational complexity, which also contributes to the research of real-time

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imaging algorithms. In short, this method is universal and can be applied to many tasks.

It is worth noting that some ship detectors in SAR nonimaging data are also introduced in [13], [14], and [15], but their detection performance is limited by the training dataset. Furthermore, the reliability of neural networks to extract ship features from nonimaging data has not been verified. Hence, owing to the rigorous demands for safety and reliability of maritime monitoring systems, these methods are unsuitable.

2) Limitation: Although a large amount of imaging calculations can be saved compared with classical CFAR, it cannot be completely avoided, especially for range compressing the raw data. Note that in this study, only scenes of spaceborne SAR monitoring ships in the vast sea area are taken into consideration. In general, there are many buildings, cars, and other strong scatterers on the ground, which are similar to ships and confuse the results. Hence, further research is needed for the scene of SAR scanning coasts and islands.

3) Optimization Points: In engineering practice, this method can be optimized to further improve system performance.

a) *Reduce Time Consumption:* In the RC-GΓD-CFAR method, each range cell is detected independently. Using distributed computing methods, the ship scattering energy of each cell can be detected in parallel. Therefore, it will further reduce the time-consuming of ship detection.

b) *Alleviate Communication Pressure:* In satellite-to-ground communication, only the range-compressed data of the detected ship needs to be transmitted to the ground. For example, in the Data-1 experiment, the total amount of data is $4192 \times 27\,600 = 115\,699\,200$, and the amount of ship data is $900 \times 50 = 45\,000$. In this way, the amount of data transmitted is reduced to 3.89×10^{-4} of the original.

VII. Conclusion

In this article, a new method of sensing ships from range-compressed SAR data is proposed, which facilitates the real-time performance of spaceborne SAR systems. First, the imaging process of SAR data is reviewed. Then, in order to suppress the sea clutter interference in the SAR range-focused domain, the RC-GΓD model is derived theoretically. Considering the estimation precision and speed, an efficient closed-form estimator for parameters is further provided. Next, the RC-GΓD-CFAR for detecting ships in range-compressed data is given. Finally, experiments are conducted on Sentinel-1 and ERS-2 SAR data. The RC-GΓD model has goodness-of-fit in the range-focused domain. The RC-GΓD-CFAR method can effectively detect ships. In future work, we will further focus on shore ship detection tasks.

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References

[1] A. Moreira, P. Prats-Iraola, M. Younis, G. Krieger, I. Hajnsek, and K. P. Papathanassiou, "A tutorial on synthetic aperture radar," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 1, pp. 6–43, Mar. 2013.

[2] G. Gao, L. Liu, L. Zhao, G. Shi, and G. Kuang, "An adaptive and fast CFAR algorithm based on automatic censoring for target detection in high-resolution SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 6, pp. 1685–1697, Jun. 2009.

[3] P. Li, Z. Ding, T. Zhang, Y. Wei, and Y. Gao, "Integrated detection and imaging algorithm for radar sparse targets via CFAR-ADMM," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5204015.

[4] M.-D. Li, X.-C. Cui, and S.-W. Chen, "Adaptive superpixel-level CFAR detector for SAR inshore dense ship detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[5] Z. Lin, K. Ji, X. Leng, and G. Kuang, "Squeeze and excitation rank faster R-CNN for ship detection in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 5, pp. 751–755, May 2019.

[6] Z. Cui, Q. Li, Z. Cao, and N. Liu, "Dense attention pyramid networks for multi-scale ship detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8983–8997, Nov. 2019.

[7] D. Li, Q. Liang, H. Liu, Q. Liu, H. Liu, and G. Liao, "A novel multidimensional domain deep learning network for SAR ship detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5203213.

[8] X. Leng, K. Ji, and G. Kuang, "Ship detection from raw SAR echo data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5207811.

[9] S. K. Joshi and S. V. Baumgartner, "Training data selection strategy for CFAR ship detection in range-compressed radar data," in *Proc. Int. Radar Conf. (RADAR)*, Sep. 2019, pp. 1–5.

[10] S. K. Joshi and S. V. Baumgartner, "Ship detection using linear and circular range-compressed airborne radar data," in *Proc. 13th Eur. Conf. Synth. Aperture Radar*, Mar. 2021, pp. 1–6.

[11] S. K. Joshi, S. V. Baumgartner, and G. Krieger, "Tracking and track management of extended targets in range-Doppler using range-compressed airborne radar data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5102720.

[12] X. Leng, J. Wang, K. Ji, and G. Kuang, "Ship detection in range-compressed SAR data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2022, pp. 2135–2138.

[13] Y. Gao, D. Li, W. Guo, and W. Yu, "SAR ship detection in range-compressed domain based on LSTM method," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2023, pp. 6422–6425.

[14] H. Zeng et al., "An incept-TextCNN model for ship target detection in SAR range-compressed domain," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[15] X. Tan, X. Leng, K. Ji, and G. Kuang, "RCShip: A dataset dedicated to ship detection in range-compressed SAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[16] G. Goldstein, "False-alarm regulation in log-normal and Weibull clutter," *IEEE Trans. Aerosp. Electron. Syst.*, vols. AES-9, no. 1, pp. 84–92, Jan. 1973.

[17] G. V. Trunk and S. F. George, "Detection of targets in non-Gaussian sea clutter," *IEEE Trans. Aerosp. Electron. Syst.*, vol. AES-6, no. 5, pp. 620–628, Sep. 1970.

[18] K. D. Ward, S. Watts, and R. J. Tough, *Sea Clutter: Scattering, the K Distribution and Radar Performance*, vol. 20. Edison, NJ, USA: IET, 2006.

[19] L. Rosenberg, S. Watts, and S. Bocquet, "Application of the K+Rayleigh distribution to high grazing angle sea-clutter," in *Proc. Int. Radar Conf.*, Oct. 2014, pp. 1–6.

[20] C. H. Gierull and I. Sikaneta, "A compound-plus-noise model for improved vessel detection in non-Gaussian SAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1444–1453, Mar. 2018.

[21] L. Rosenberg and S. Angelliaume, "Characterisation of the tri-modal discrete sea clutter model," in *Proc. Int. Conf. Radar (RADAR)*, Aug. 2018, pp. 1–6.

[22] S. Angelliaume, L. Rosenberg, and M. Ritchie, "Modeling the amplitude distribution of radar sea clutter," *Remote Sens.*, vol. 11, no. 3, p. 319, Feb. 2019.

[23] G. Gao, S. Gao, K. Ouyang, J. He, and G. Li, "Scheme for characterizing clutter statistics in SAR amplitude images by combining two parametric models," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5636–5646, Oct. 2018.

[24] T. Li, D. Peng, Z. Chen, and B. Guo, "Superpixel-level CFAR detector based on truncated gamma distribution for SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 8, pp. 1421–1425, Aug. 2021.

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[25] G. Gao, K. Ouyang, Y. Luo, S. Liang, and S. Zhou, "Scheme of parameter estimation for generalized gamma distribution and its application to ship detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 3, pp. 1812–1832, Mar. 2017.

[26] W. Zhou, J. Xie, B. Zhang, and G. Li, "Maximum likelihood detector in gamma-distributed sea clutter," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 11, pp. 1705–1709, Nov. 2018.

[27] H.-C. Li, W. Hong, Y.-R. Wu, and P.-Z. Fan, "On the empirical-statistical modeling of SAR images with generalized gamma distribution," *IEEE J. Sel. Topics Signal Process.*, vol. 5, no. 3, pp. 386–397, Jun. 2011.

[28] X. Qin, S. Zhou, H. Zou, and G. Gao, "A CFAR detection algorithm for generalized gamma distributed background in high-resolution SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 4, pp. 806–810, Jul. 2013.

[29] J. Li, M. Zhang, W. Fan, and D. Nie, "Facet-based investigation on microwave backscattering from sea surface with breaking waves: Sea spikes and SAR imaging," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 4, pp. 2313–2325, Apr. 2017.

[30] Y. Chen, Y. Sun, and Q. Liu, "Parametric iterative soft thresholding algorithm for refocusing of moving targets in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5229109.

[31] I. Cumming and F. Wong, "Digital processing of synthetic aperture radar data: Algorithms and implementation," *Artech House*, vol. 1, no. 3, pp. 108–110, 2005.

[32] S. Gao and H. Liu, "Performance comparison of statistical models for characterizing sea clutter and ship CFAR detection in SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 7414–7430, 2022.

[33] R. Gu, S.-X. Zhang, Y. Zou, and D. Yu, "Towards unified all-neural beamforming for time and frequency domain speech separation," *IEEE/ACM Trans. Audio, Speech, Language Process.*, vol. 31, pp. 849–862, 2023.

[34] B. Gold, N. Morgan, and D. Ellis, *Speech and Audio Signal Processing: Processing and Perception of Speech and Music*. Hoboken, NJ, USA: Wiley, 2011.

[35] C. M. Bishop and N. M. Nasrabadi, *Pattern Recognition and Machine Learning*, vol. 4. New York, NY, USA: Springer, 2006.

[36] J.-M. Nicolas and S. N. Anfinsen, "Introduction to second kind statistics: Application of log-moments and log-cumulants to the analysis of radar image distributions," *Trait. Signal*, vol. 19, no. 3, pp. 139–167, 2002.

[37] T. Liu, Z. Yang, A. Marino, G. Gao, and J. Yang, "Robust CFAR detector based on truncated statistics for polarimetric synthetic aperture radar," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 9, pp. 6731–6747, Sep. 2020.

[38] M. Dohler and M. Arndt, "Inverse incomplete gamma function and its application," *Electron. Lett.*, vol. 42, no. 1, pp. 35–36, 2006.

[39] ASF DAAC. (Dec. 7, 2014). *Dataset: ERS-2, ESA 1993*. [Online]. Available: https://asf.alaska.edu/datasets/daac/ers-2/

[40] *Sentinel-1 SAR Space Packet Protocol Data Unit*, document S1-IF-ASD-PL-0007 no. 13, 2015. [Online]. Available: https://sentinels.copernicus.eu/web/sentinel/user-guides/sentinel-1-sar/document-library

[41] *Sentinel-1 Level-0 Product Format Specifications*, document S1PD.SP.00110.ASTR no. 1, 2012. [Online]. Available: https://sentinels.copernicus.eu/web/sentinel/user-guides/sentinel-1-sar/document-library

[42] *Sentinel-1 Level-1 Detailed Algorithm Definition*, document DI-MPC-IPFDPM Issue 2.5, 2022. [Online]. Available: https://sentinels.copernicus.eu/web/sentinel/user-guides/sentinel-1-sar/document-library

[43] A. MartÌnez-UsÓMartinez-Uso, F. Pla, J. M. Sotoca, and P. GarcÌa-Sevilla, "Clustering-based hyperspectral band selection using information measures," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 12, pp. 4158–4171, Dec. 2007.

[44] B. Dan, M. Li, T. Tang, X. Qi, Z. Zhu, and Y. Ouyang, "Spatial–temporal stochastic resonance model for dim-small target detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.



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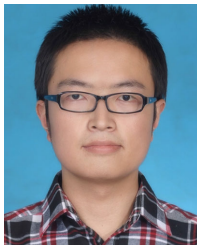
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